

# RT-IOT2022



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# RT-IOT2022 Dataset

| Descriptor     | Value    |
|----------------|----------|
| Feature Count  | 83       |
| Size           | 53MBytes |
| Row Count      | 123,118  |
| Missing Values | No       |

# Problem

Can a Decision Tree classify traffic based on the training data in the dataset?

# Makefile

- Supports rapid iteration
- Checks for proper configuration
- Stages and retrieves data from HDFS
- Proper dependency checking to avoid re-execution

```
$ make all
```

```
$ make clean
```

# Preprocessing

- Remove row\_id field
- Rename fields with “\_” in them
- StringIndex text fields and the label

# TargetEncoding

- Encodes a {column, target} pair into a likelihood
- Allows categorical fields (text, port numbers) to be standardized and passed to PCA
- Converts a categorical field to the range  $[0, 1]$

# Cluster vs Local

- Single baseline pass on a laptop took 15s
- Single baseline pass in the cluster took 75s
- Cluster was unable to run 10 model trainings in parallel

# Conclusion

- Spark is a tool used only when it is “necessary”
- DecisionTree doesn't work well with this dataset
- A pipeline that tests all model types would be useful